Response to Reviewer 2

We sincerely thank the reviewer for their thoughtful feedback, their positive assessment of the manuscript's readability, and for recognizing the significance of applying MAB-based model selection to the financial sector. We agree that ensuring generalizability and transparency is paramount. Below, we address your concerns point-by-point, detailing the specific updates made to the manuscript.

General Comments

Response: To address your overarching concerns regarding literature context, data transparency, and reproducibility, we have made the following specific updates to the manuscript:

- **Contextualizing with Recent Literature:** We appreciate the reviewer pointing out the recent successes of MAB approaches in other non-stationary environments. In **Section 1 (Introduction)**, we have added a discussion and cited the suggested recent works (Pandian et al., 2025; Wilson et al., 2024) to better position our study within this broader, successful trend of dynamic ensemble selection.
- **Detailed Data Description:** We agree that the data description was previously insufficient. We have expanded **Section 3.1 (Environment Construction)** to explicitly detail our dataset. The text now clearly states that we utilized daily closing prices sourced via the Yahoo Finance API, with a data period starting from January 1, 2000, for all available assets. Furthermore, to clarify the evaluation in Table 1, we explicitly state that a 0% risk-free rate was assumed for the calculation of all risk-adjusted performance metrics.
- **Code and Data Availability:** To ensure full transparency and reproducibility of our analysis, we have added a dedicated **"Data and Code Availability"** section at the end of the manuscript. This section provides a direct link to a public GitHub repository containing all the Python code used for data collection, environment construction, and model evaluation.

Specific Comments

0.1 Logit-type Selection

Reviewer's Comment: *In this study, a Softmax policy (logit-type selection) is introduced for model selection during training, and a Greedy policy during testing... If the essence of MAB lies in the balance between exploitation and exploration, it would be desirable to examine how performance changes when introducing full-fledged MAB methods that explicitly handle the exploration component, such as UCB or Thompson Sampling, rather than limiting it a priori to a logit model.*

Response: We sincerely thank the reviewer for this excellent and highly constructive observation. We entirely agree that testing full-fledged MAB algorithms that explicitly model the exploration-exploitation trade-off provides a much stronger foundation for the paper.

To address this, we have significantly expanded both our methodology and our experimental validation:

- In **Sections 2.2 and 2.3 (Background)**, we have added detailed theoretical descriptions of two prominent Action-Value policies: Upper Confidence Bound (UCB) and Thompson Sampling (TS), distinguishing their explicit exploration mechanisms from the policy-learning Softmax approach.
- We ran comprehensive new experiments incorporating UCB and Thompson Sampling across all five asset classes. The results of these experiments are now detailed in **Section 4 (Results and Performance Analysis)** and summarized in **Table 1**.

Crucially, the new empirical results confirm the reviewer's intuition perfectly. As discussed in the revised text, the UCB policy emerges as the dominant strategy, consistently outperforming the simple logit-type Softmax model in terms of Annualized Return and Sharpe Ratio across almost all asset classes. This addition greatly strengthens our conclusions regarding the necessity of explicit exploration mechanisms in non-stationary financial environments.

0.2 MAB Setting

On SGA, Logit Selection, and Variance: The reviewer makes an exceptionally astute observation regarding the theoretical underpinnings of the Softmax (logit) selection. Our initial motivation for optimizing θ via SGA based on expected returns was to explicitly mimic the Policy Gradient theorem used in full Reinforcement Learning (e.g., REINFORCE), treating the CMAB as a single-step MDP where the policy is directly parameterized and optimized for expected reward. However, we completely agree with the reviewer's critique from a strict MAB perspective: the logit model implicitly assumes errors drawn from a Type I Extreme Value distribution, where the response coefficients inherently dictate the scale (variance) of exploration. By mirroring standard policy gradients, our formulation failed to explicitly map this scale parameter to the actual variance or uncertainty of the models' predictive performance. It was precisely to address this theoretical limitation that we expanded our methodology and experiments to include **LinUCB** and **Thompson Sampling** (detailed in our response to Comment 1), which correctly align the exploration scale with statistical confidence.

On Risk in Context ([Huo and Fu, 2017]): We appreciate the reference to Huo and Fu [2017], which demonstrates risk-aware MABs for continuous portfolio allocation. We have added a new paragraph at the end of **Section 3.3 (Context Construction)** explicitly citing this work and clarifying why variance was excluded from our context vectors. As we now detail in the text, our contexts capture the discrepancy between predicted and actual returns for each specific arm. If we were to incorporate risk via the variance of the underlying asset, this metric would be identical for all arms at any given time step. Consequently, this shared variance would act merely as a global bias term rather than providing the bandit with arm-specific discriminative power to aid in model selection. Therefore, we deliberately focused our context on model-specific predictive accuracy.

0.3 Dataset for Model Evaluation

Reviewer's Comment: *The S&P 500 index is used here for model evaluation, but verification using several datasets is desired to demonstrate the robustness of the proposed model's performance. For example, additional verification using datasets commonly used in financial model evaluation, such as the French 25, 48, and 100 portfolios or ETF139, and the presentation of analysis results with different evaluation periods would make the author's claims more persuasive.*

Response: We completely agree with the reviewer that evaluating the framework solely on the S&P 500 index was insufficient to prove the robustness and generalizability of the proposed method.

To address this, we have massively expanded our experimental evaluation. Rather than limiting the expansion to other equity portfolios, we opted to test the model across entirely different market microstructures to rigorously stress-test its generalizability. As detailed in the revised **Section 3.1 (Environment Construction)** and **Section 4 (Results and Performance Analysis)**, we now evaluate the MAB policies across five diverse asset classes:

- **SPY** (Broad Market Equities)
- **BTC-USD** (Cryptocurrency)
- **TLT** (Sovereign Fixed Income / Bonds)
- **EURUSD=X** (Foreign Exchange)
- **GC=F** (Commodities / Gold)

Furthermore, to address the reviewer's request regarding evaluation periods, we clarified in **Section 3.1** that our testing period utilizes daily data starting from January 1, 2000 (for all assets where such history is available). This extensive 24-year period naturally encompasses multiple, highly distinct market regimes, including the recovery from the Dot-Com bubble, the 2008 Great Financial Crisis, the 2020 COVID-19 crash, and subsequent inflationary periods.

The results of these extensive new experiments are now summarized in **Table 1** and visualized in the new **Figure 5**. They consistently demonstrate that the explicitly exploratory MAB algorithms (particularly UCB) maintain their robust outperformance across these diverse datasets and time periods.

0.4 Model Verification (Overfitting and Synthetic Data)

Reviewer's Comment: *Since there is a concern about overfitting, I strongly suggest considering simulation-based verification as follows. I propose creating several data series using standard models of financial time series analysis, such as the EGARCH model, where volatility has strong autocorrelation but returns themselves have no predictability, and verifying whether the model shows high performance on these completely unpredictable series...*

Response: We sincerely thank the reviewer for this rigorous and highly practical suggestion. A "placebo test" on pure noise is indeed the gold standard for proving that a complex financial machine learning pipeline is not hallucinating predictability.

Following your recommendation, we have added a new subsection, **Section 4.3 (Robustness Check: Synthetic Data Simulation)**. We simulated a synthetic S&P 500-like asset using a GARCH(1,1) process. This dataset was calibrated to feature realistic volatility clustering and a constant upward drift, but the directional variations around that drift were fundamentally unforecastable noise.

We then passed this synthetic dataset through our entire pipeline—training the heterogeneous predictor pool and applying the CMAB selection policy. As hypothesized, the CMAB framework failed to generate any excess returns. Because there were no genuine structural inefficiencies to exploit, the dynamic switching mechanism of the MAB suffered from whipsaw against the random noise, ultimately underperforming the static, long-biased Ensemble Mean baseline.

This experiment conclusively demonstrates what the reviewer hypothesized: our proposed framework does not overfit to pure noise or autocorrelated volatility. It confirms that the performance observed on the real-world datasets is the result of the MAB successfully dynamically selecting models that capture actual, underlying market predictability. We are very grateful for this suggestion, as it substantially strengthens the credibility of our empirical results.

0.5 Selection of Base Models

Reviewer's Comment: *Here, an ensemble of only simple Deep Multi-Layer Perceptrons (DMLPs) with two hidden layers is considered, but please explain/verify the validity of limiting it to simple DMLPs... there are already studies that sequentially select multiple trading models, such as Deep Q-Networks, using MAB [Zhu et al., 2019, Zeng et al., 2023]. What is the advantage of this study when compared to those?*

Response: We sincerely thank the reviewer for this highly constructive comment. We address the two points—the limitation of DMLPs and the comparison to RL-based MAB frameworks—as follows:

- **Expanding beyond simple DMLPs:** We entirely agreed with the reviewer's concern that limiting the base models to simple DMLPs artificially restricted the capability of the bandit framework. Consequently, we significantly expanded our methodology. As detailed in the revised **Section 3.2.1 (Impact of Architectural Heterogeneity)**, our predictor pool now consists of a heterogeneous mix of architectures, including DMLPs, Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs). As shown in our new empirical results across all five datasets, this structural heterogeneity is exactly what allows the explicitly exploratory MAB policies (like UCB) to shine, dynamically exploiting the diverse inductive biases of the different networks depending on the localized market regime.
- **Comparison to MABs over Deep RL Agents (Zhu et al. / Zeng et al.):** We are grateful for these excellent references, which we have now explicitly discussed and cited in our revised manuscript. Studies like Zhu et al. [2019] and Zeng et al. [2023] utilize MABs to select between full-fledged Reinforcement Learning (RL) agents (e.g., Deep Q-Networks) that map states directly to trading actions.

  The distinct advantage of our study lies in operating strictly at the **supervised prediction layer**. By utilizing CMABs to select among supervised forecasters rather than RL agents, we decouple the variance of localized predictability from the severe instability, delayed reward mechanisms, and complex credit-assignment challenges inherent to multi-agent Deep RL. Our framework isolates pure structural inductive biases (e.g., an RNN's temporal memory vs. a CNN's feature extraction), resulting in an architecture that is vastly more computationally efficient, highly interpretable, and directly minimizes immediate predictive error.

0.6 Results in Table 1 (Sharpe Ratio, Risk, and Sub-periods)

Reviewer's Comment: *Regarding the Sharpe ratio, assuming a 3-month T-bill as the risk-free rate for daily data, it is generally around 0.5 for the period 2004-2024. This value significantly exceeds the result in Table 1 (about 0.27). Please provide detailed information regarding this discrepancy. From the viewpoint of performance stability, how is the model in this study evaluated? Including a description where the data period is divided into several characteristic periods would deepen the reader's understanding...*

Response: We sincerely thank the reviewer for this excellent, finance-grounded perspective. We completely agree that the initial Sharpe ratio of 0.27 presented in the original manuscript was anomalous compared to the historical Buy-and-Hold (B&H) Sharpe ratio of the S&P 500 (≈ 0.50).

- **The Sharpe Ratio Discrepancy Resolved:** During our extensive revisions and code refactoring prompted by your comments, we identified and corrected a minor calculation error in our pipeline regarding how the percentage change (`pct_change`) of the predictions was computed. Correcting this implementation detail, combined with the introduction of the explicitly variance-aware exploration policies (**LinUCB** and **Thompson Sampling** discussed in Comment 1), completely resolves this discrepancy. As shown in our newly revised **Table 1**, the Sharpe ratio for the SPY dataset under the UCB policy is now approximately **0.65**, which successfully *exceeds* the historical B&H market benchmark.
- **Sub-period Analysis and Risk Stability:** Regarding how the model evaluates on risk stability: this strong risk-adjusted outperformance is achieved because the zero-beta Long/Short framework provides downside protection during structural breaks. Our overarching evaluation naturally encompasses the exact sub-periods the reviewer mentions. During severe drawdowns like the 2008 Lehman Shock, the Long/Short CMAB rapidly shifts weight to models capturing the downward trend, mitigating Maximum Drawdown (MDD) and generating positive alpha while Long-Only indices collapse. Conversely, during aggressive, low-volatility secular bull markets (such as the post-2020 AI rally), the strategy's exploratory short positions naturally act as a slight drag on absolute returns relative to a passive benchmark.

Ultimately, the strategy's ability to aggressively cut tail-risk during crashes is what drives the full-period Sharpe ratio up to 0.60. Because our full out-of-sample plots visually capture this dynamic behavior and survival across varying regimes, we opted to keep the main manuscript focused on the broad multi-asset evaluation rather than breaking the text into fragmented sub-period analyses. We are highly grateful for this comment, as it prompted the deep dive that ultimately corrected and vastly improved our reported empirical results!

0.7 Descriptions (Notation and Table 1 Details)

Reviewer's Comment: *I was slightly confused about the subscript t in Section 2.3. First, what is the difference between θ_a and $\theta_{a,t}$? Also, is the reversal of subscripts in $x_{t,a}$ and $\theta_{a,t}$ due to the difference between column and row vectors? Please provide detailed information regarding the evaluation in Table 1...*

Response: We thank the reviewer for their meticulous reading of the manuscript. Clear notation and reproducible metric definitions are vital, and we appreciate the opportunity to correct these ambiguities.

- **Subscript Notation (Section 2.3):** We apologize for the notational inconsistency. The subscript reversal ($x_{t,a}$ vs $\theta_{a,t}$) was not due to a column/row vector distinction, but was simply a typographical oversight. We have now standardized the notation throughout the paper to strictly use $\{t, a\}$ (e.g., $x_{t,a}$ and $\theta_{t,a}$).
- **Table 1 Evaluation Details:** We entirely agree that the exact parameters of the financial metrics must be transparent. We have significantly expanded the caption of **Table 1** and the text in **Section 4** to include these precise definitions. Specifically, we clarify that the evaluation utilizes **daily return frequency** over the overarching period of January 2000 to January 2024. Furthermore, the annualized Sharpe ratio is calculated assuming 252 trading days and a **risk-free rate of 0%**, which is the standard convention for evaluating the pure predictive alpha of zero-beta Long/Short daily trading strategies.